

Network methods to study plant xylem

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Networks: what and why?

Network: a model for connections and interactions

- Internet, public transport, social networks

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Nodes: network's basic elements

- Web pages, stops, people
- In tree xylem, vessels (and possibly tracheids and fibres)

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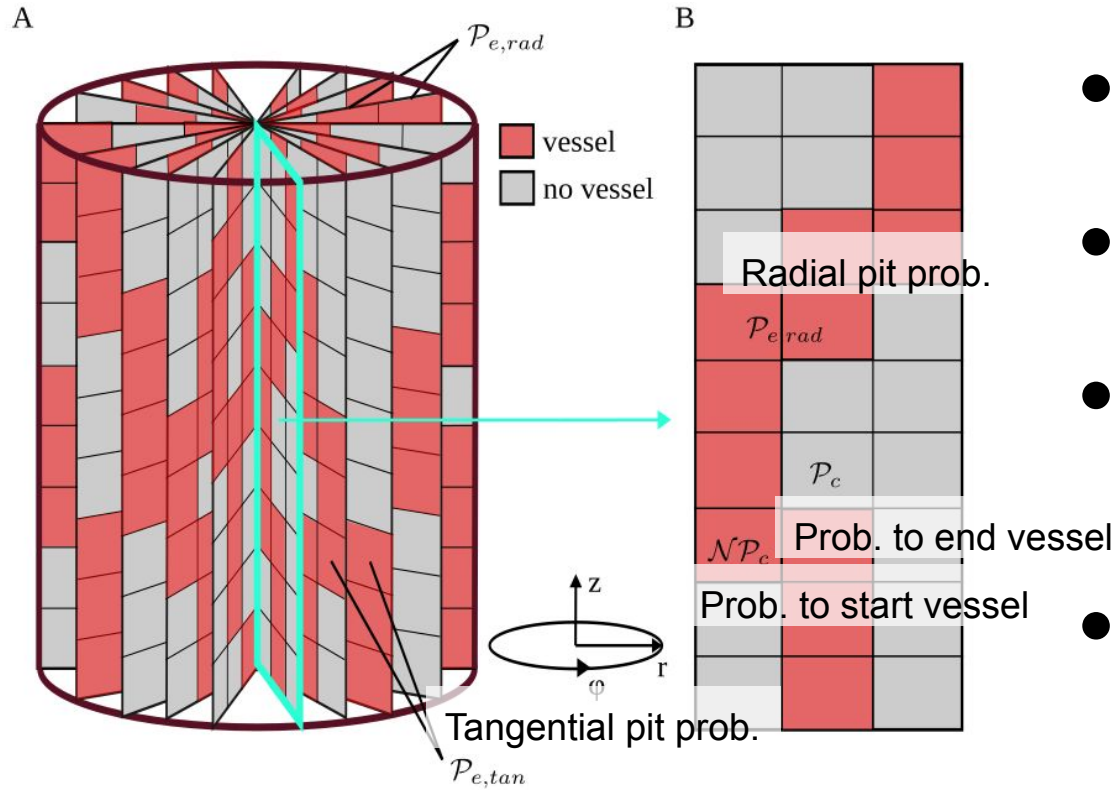
- Web pages, stops, people
- In tree xylem, vessels (and possibly tracheids and fibres)

Links: connections between nodes

- Web links, transport lines, social relationships
- In tree xylem, pits

Stochastic xylem network model

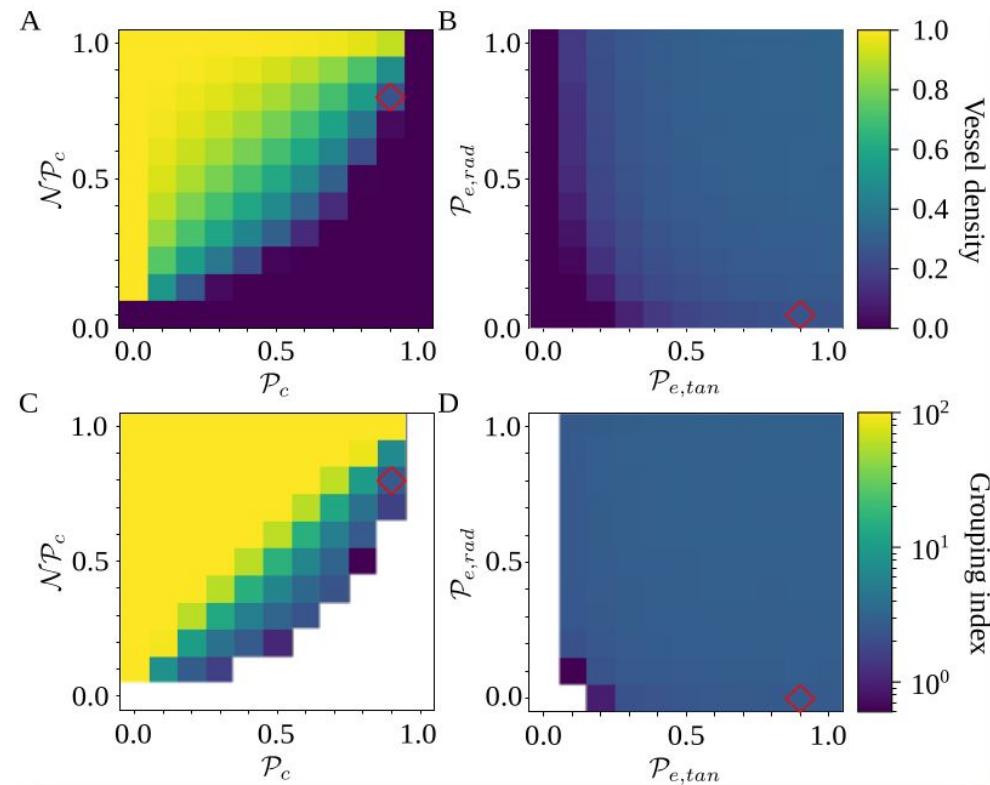
- Original: Loepfe et al., 2007; Mrad et al., 2018, 2021
- Involved in present study: S. Jansen, L. de Melo Silva, P. Kiuru, M. Held, A. Lintunen, A. Laurén



- Xylem modelled as a cylindrical 3D grid
- Four probability parameters
- Network nodes: vessels, network links: intervessel pits
- A testbench for studying embolism spreading

Stochastic xylem network model

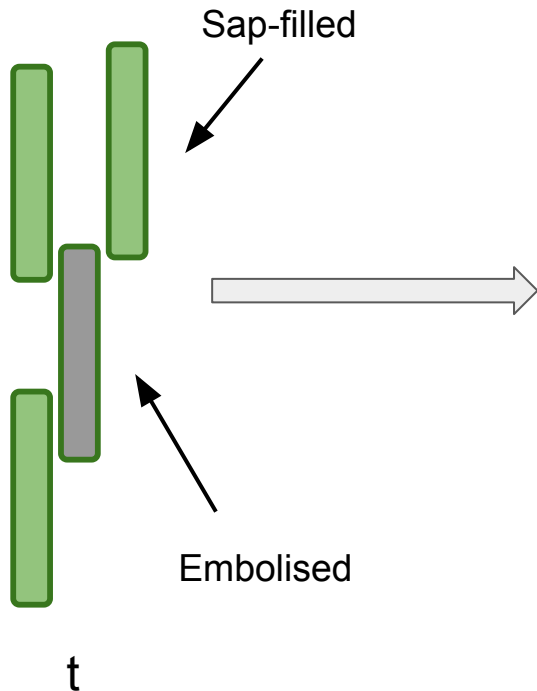
Original: Loepfe et al., 2007; Mrad et al., 2018, 2021



- Originally build for *Acer glabrum*
- Re-parameterized for *Betula pendula* based on vessel density & grouping index
- Next step: comparison to CT-based vessel network reconstructions

Models for embolism spreading

Run on top of the xylem network

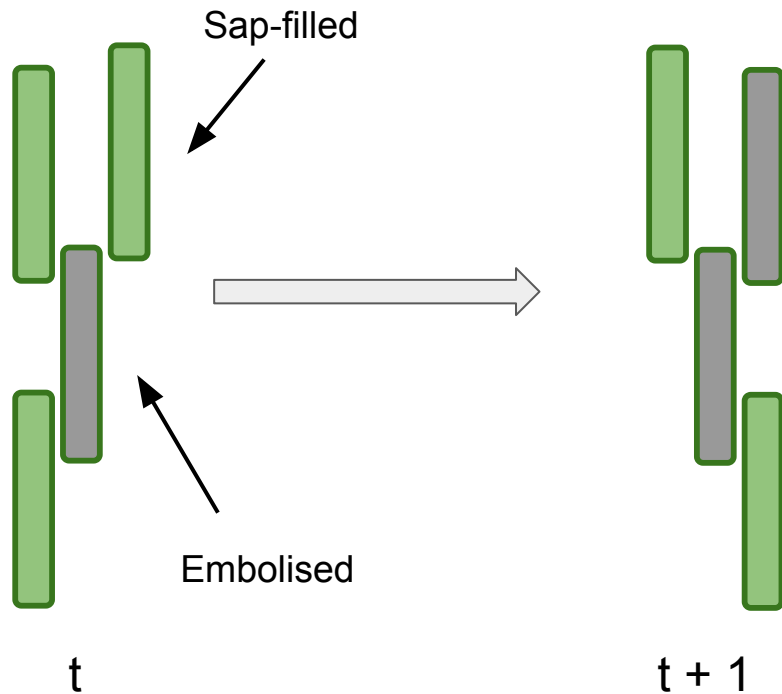


Physiological model:

- Embolism spreads if sap pressure exceeds bubble propagation pressure (BPP)
- Pores modelled as tunnels consisting of several constrictions (Kaack et al., 2021)
- Smallest constriction of each pores R_{min} , max across all pores $R_{min,max}$
- $BPP = k\gamma 2\cos(\alpha)/R_{min,max}$

Models for embolism spreading

Run on top of the xylem network

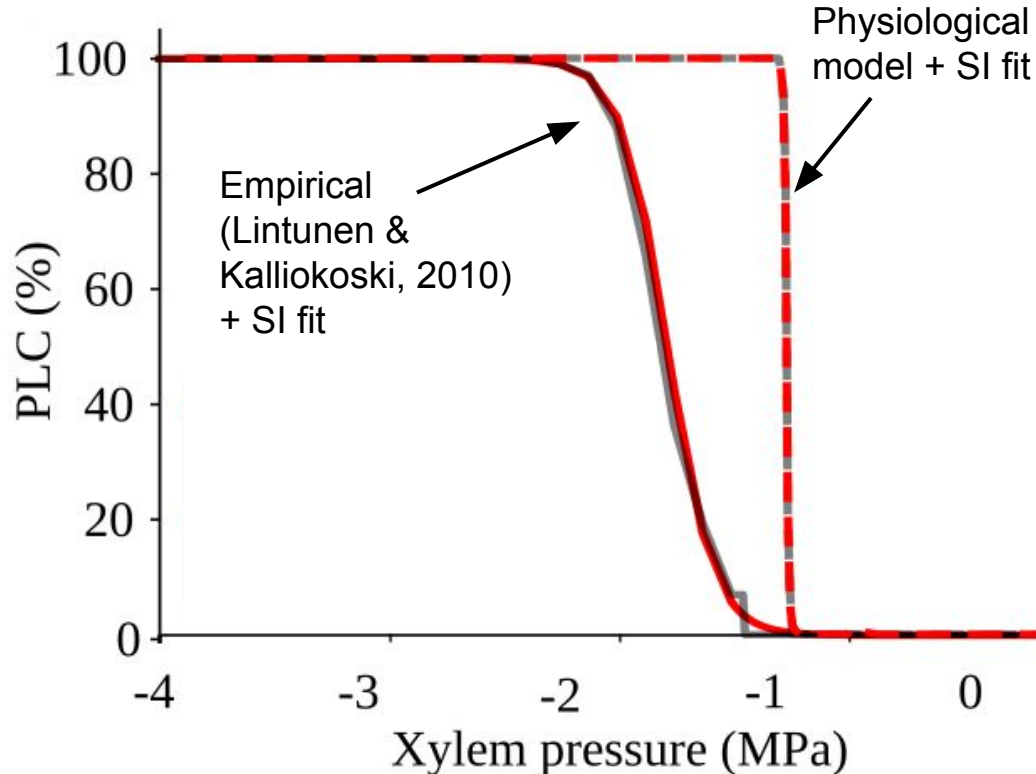


Susceptible-Infected (SI) model:

- A stochastic compartment model (Pastor-Satorras et al., 2015)
- Embolism spreads with probability β
- Spreading probability is easy to fit against empirical data
- Roth-Nebelsick (2019) and Torres-Ruiz et al. (2016) applied compartment models on xylem ignoring connectivity structure

Models for embolism spreading

Run on top of the xylem network

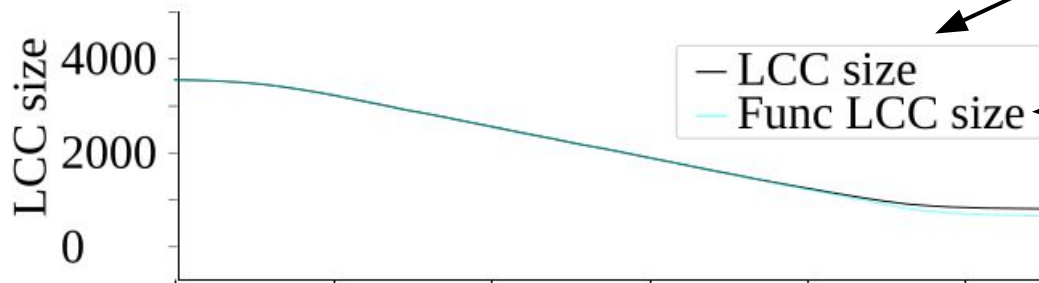


Advantages of the SI model

- Uses a single free parameter
- Easily fittable to empirical vulnerability curves
- Allows monitoring of xylem properties during spreading
- Commonly used in network science => allows comparison to other spreading phenomena

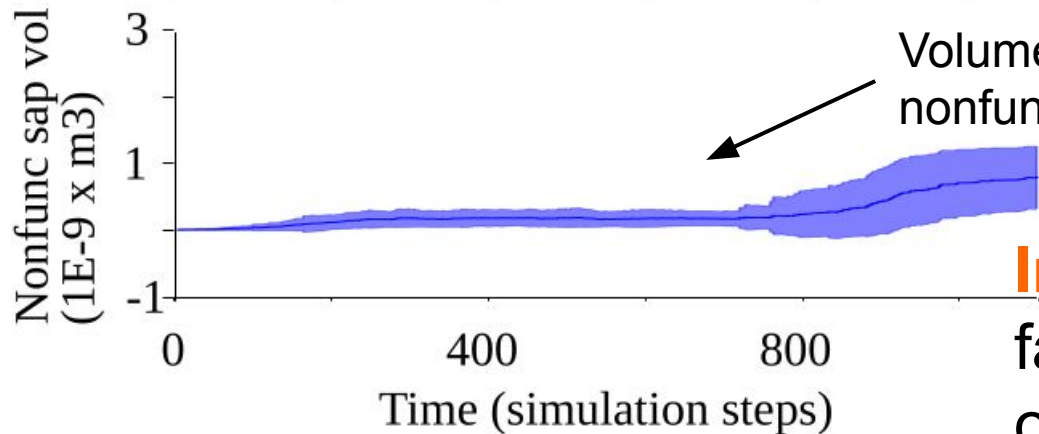
Models for embolism spreading

SI embolism spreading at P_{50}



Set of nodes connected by links

Connected to both inlet and outlet



Volume of sap in nonfunctional vessels

Interpretation: hydraulic failure is caused by both loss of sap and network breakdown

Conclusions

- Networks model connections and interactions
- One potential application target: modelling embolism spreading
 - Stochastic xylem network model creates testbenches
 - SI spreading model replicates empirical vulnerability curves building on a single free parameter
- **Future directions:**
 - Comparison of simulated and reconstructed networks
 - Scaling of embolism spreading with vessel dimensions
 - Origin of embolism: including bubble expansion in the model
 - Combining vessels, tracheids and fibres: multilayer networks
 - Other?

References

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- Lintunen, A., Kalliokoski, T.,** 2010. The effect of tree architecture on conduit diameter and frequency from small distal roots to branch tips in *Betula pendula*, *Picea abies* and *Pinus sylvestris*. *Tree Physiology* 30(11).
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Thank you!

Questions, comments?

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Slides:

https://github.com/onerva-korhonen/presentations/blob/master/ulm_171125.pdf